



EXECUTION PLAN

VIRTUAL CONTROL

VMWARE CLOUD FOUNDATION SOLUTIONS

vCLS Retreat + MM Retry + esxi01 Disk Fix

Verified procedure to repair esxi01's sata0:4 `IsSSD=false` misclassification after the 2026-05-14 MM Full Data Migration failure. Uses vCLS Retreat Mode to clear the placement-pinned blocker, then follows the lab's documented vSAN OSA disk-group rebuild path.

VSAN OSA

VCLS RETREAT

MAINTENANCE MODE

VIRTUALSSD

CAPACITYFLASH

VCF 9.0.1

VMware Cloud Foundation

vCLS Retreat + MM Retry + esxi01 Disk Fix

Prepared by: Virtual Control LLC **Date:** May 15, 2026 **Document Version:** 1.0 **Classification:** Internal — Lab Operations **Target host:** esxi01.lab.local **Target disk:** t10.ATA_____VMware_Virtual_SATA_Hard_Drive_____04000000000000000001 (sata0:4, 100 GB addon disk, esxi01.lab.addon.vmdk) **Companion docs:** [vSAN-OSA-Disk-Group-Rebuild-Handbook-2026-05-08.md](#) — the procedure pattern this plan adapts

1. Context

1.1 What happened May 14

Yesterday at 22:20 UTC the operator initiated MM with Full Data Migration on esxi01 to perform the documented vSAN OSA disk-group rebuild for sata0:4 (IsSSD=false). MM failed at ~30% with:

General vSAN error. Failed to enter maintenance mode in the current vSAN data migration mode due to insufficient nodes or disks in the cluster. Objects Evacuated: 57 of 159, Data Evacuated: 115985 MB of 397924 MB

Concurrent vSphere events showed esxi02 had a brief connection loss at 22:30:06 UTC during the MM operation:

22:30:06 — HostConnectionLostEvent on esxi02.lab.local 22:30:10 — Cannot synchronize host esxi02.lab.local 22:21–22:31 — Multiple IOFilter VASA Provider registration failures on esxi02 (port 1443)

With esxi02 transiently offline during MM, vSAN saw only 2 active hosts (esxi03 + esxi04) for FTT=1 evacuation — insufficient.

1.2 Suspected secondary blocker — vCLS placement

vCLS-4efd4d56-... was placed on esxi01 at the time of MM. vCLS VMs are vSphere-managed and DRS cannot freely migrate them — they are governed by anti-affinity rules at the cluster level. The Broadcom-documented mechanism to retire vCLS VMs is **Retreat Mode**, via the cluster advanced setting `config.vcls.clusters.<cluster-id>.enabled = False`.

1.3 What changed before this retry

- All 8 cluster snapshots consolidated (collector, sddc-manager, vcenter, fleet, vcf-ops) — vSAN object count dropped 193 → 149
- Free space recovered 41% → 49.3% (1067 → 1281 GB)
- NSX cold-migrated to esxi04 with re-bound NIC (was on orphan vDS UUID — now on pg-vm-mgmt on the live vDS)
- SDDC Manager SSO trust repaired (vCenter leaf + VMCA root imported to both truststores per separate procedure)

2. Pre-Execution Gates

All four must pass before Step 1:

#	Gate	Verify with
G1	Resync = 0 GB / 0 objects	<code>esxi02 esxcli vsan debug resync summary get</code>
G2	No snapshots in cluster	<code>Get-VM \ Get-Snapshot</code> returns empty
G3	esxi01 has no VMs except vCLS	<code>Get-VMHost esxi01 \ Get-VM</code> shows only <code>vCLS-*</code>
G4	NSX VIP responds HTTP 200	<code>Test-NetConnection nsx-vip.lab.local:443</code> + HTTP probe
G5	All hosts Connected, none in MM	<code>Get-VMHost \ Select ConnectionState, InMM</code>

3. Step-by-Step Procedure

Step 1 — Enable vCLS Retreat Mode [BROADCOM-DOC: KB 80472]

```
$cl = Get-Cluster vcenter-cl01
$key = "config.vcls.clusters.$($cl.Id.Replace('ClusterComputeResource-', '')).enabled"
# Verified value for this cluster: config.vcls.clusters.domain-cl0.enabled

$existing = Get-AdvancedSetting -Entity $vi -Name $key -ErrorAction SilentlyContinue
if ($existing) {
    $existing | Set-AdvancedSetting -Value 'False' -Confirm:$false
} else {
    New-AdvancedSetting -Entity $vi -Name $key -Value 'False' -Type ConfigParam -Confirm:$false
}
```

Step 2 — Wait for vCLS VMs to be retired

```
for ($i=1; $i -le 10; $i++) {
    $remaining = Get-VM -Name 'vCLS*' -ErrorAction SilentlyContinue
    if (-not $remaining) { "vCLS retirement complete after $($i*30) sec"; break }
    " poll $i: $($remaining.Count) vCLS VMs still present"
    Start-Sleep 30
}
```

Gate to Step 3: Zero `vCLS-*` VMs in inventory.

Step 3 — Enter MM with Full Data Migration on esxi01 [BROADCOM-DOC: KB 315532]

```
Set-VMHost -VMHost (Get-VMHost esxi01.lab.local) -State Maintenance -Evacuate -Confirm:$false
```

Or via vCenter UI: right-click esxi01 → Maintenance Mode → Enter Maintenance Mode → **Full Data Migration**.

Wait ~15-30 min for completion. Monitor in vCenter Recent Tasks.

Step 4 — Verify MM and remove disk group [\[BROADCOM-DOC: KB 315532\]](#)

```
(Get-VMHost esxi01.lab.local).ExtensionData.Runtime.InMaintenanceMode # expect True

$exc01 = Get-EsxCli -VMHost (Get-VMHost esxi01.lab.local) -V2
$args = $exc01.vsan.storage.remove.CreateArgs()
$args.uuid = '521f8f5c-9912-c37b-461d-7b0b94e01e0c'
$args.evacuationmode = 'noAction'
$exc01.vsan.storage.remove.Invoke($args)
```

Step 5 — Power off esxi01 nested VM in VMware Workstation (operator)

In Workstation: right-click "Clone of esxi01.lab.local" → Power → **Shut Down Guest** (graceful).

Step 6 — Edit `.vmx` (operator) [\[VMware KB 1006827\]](#)

Open `D:\VMs\esxi01.lab.local\Clone of esxi01.lab.local.vmx` in a text editor.

Add this line in the SATA section, immediately after `sata0:3.virtualSSD = "1"` (if present) or anywhere in the disk block:

```
sata0:4.virtualSSD = "1"
```

Save and close.

Step 7 — Power on esxi01 nested VM (operator)

In Workstation: power on. Wait 3-5 min for ESXi to fully boot and rejoin vCenter.

Step 8 — Verify IsSSD=true on sata0:4

```
$exc01 = Get-EsxCli -VMHost (Get-VMHost esxi01.lab.local) -V2
$exc01.vsan.storage.list.Invoke() | Select Device, IsSSD, IsCapacityTier, InCMMDS
```

Gate to Step 9: disk ending in `04000000000000000001` shows `IsSSD=true`.

Step 9 — Apply capacityFlash tag [\[Broadcom TechDocs – Mark Flash Devices as Capacity Using ESXCLI\]](#)

```
$args = $exc01.vsan.storage.tag.add.CreateArgs()
$args.disk = 't10.ATA_____VMware_Virtual_SATA_Hard_Drive_____04000000000000000001'
$args.tag = 'capacityFlash'
$exc01.vsan.storage.tag.add.Invoke($args)
```

Step 10 — Recreate disk group [\[BROADCOM-DOC: KB 315532\]](#)

```
$args = $exc01.vsan.storage.add.CreateArgs()
$args.ssd = 't10.ATA_____VMware_Virtual_SATA_Hard_Drive_____000000000000000001'
$args.disks = @(
    't10.ATA_____VMware_Virtual_SATA_Hard_Drive_____02000000000000000001',
    't10.ATA_____VMware_Virtual_SATA_Hard_Drive_____03000000000000000001',
    't10.ATA_____VMware_Virtual_SATA_Hard_Drive_____04000000000000000001'
)
$exc01.vsan.storage.add.Invoke($args)
```

Step 11 — Verify both `IsSSD=1` and `IsCapacityFlash=1`

```
$exc01.vsan.storage.list.Invoke() | Where { $_.Device -match '040000000000000001' } | Select Device,
IsSSD, IsCapacityTier, InCMMDS, IsMounted
```

Gate to Step 12: `IsSSD=true`, `IsCapacityTier=true`, `InCMMDS=true`, `IsMounted=true`.

Step 12 — Exit MM

```
Set-VMHost -VMHost (Get-VMHost esxi01.lab.local) -State Connected -Confirm:$false
```

Step 13 — Disable Retreat Mode (recreate vCLS)

```
$existing = Get-AdvancedSetting -Entity $vi -Name $key
$existing | Set-AdvancedSetting -Value 'True' -Confirm:$false
```

vSphere recreates vCLS VMs within 1-3 min on healthy hosts.

Step 14 — Wait for resync to reach zero

```
for ($i=1; $i -le 30; $i++) {
    $rs = $exc.vsan.debug.resync.summary.get.Invoke()
    " poll $i: $($rs.TotalGBLeftToResync) GB / $($rs.TotalNumberOfResyncingObjects) objects"
    if ($rs.TotalGBLeftToResync -eq 0 -and $rs.TotalNumberOfResyncingObjects -eq 0) { "Resync complete";
break }
    Start-Sleep 60
}
```

4. Source Authority

Step(s)	Source
1, 2, 13	[BROADCOM-DOC: KB 80472] — Putting vSphere Cluster Services (vCLS) in Retreat Mode
3, 4, 10	[BROADCOM-DOC: KB 315532] — How to manually remove and recreate a vSAN disk group
6	[VMware KB 1006827] — Configuring SSD virtual disks for VMware Workstation/Fusion

Step(s)	Source
9, 11	[Broadcom TechDocs] — Mark Flash Devices as Capacity Using ESXCLI
Pattern	vSAN-OSA-Disk-Group-Rebuild-Handbook-2026-05-08.md §4 (lab-verified on esxi02/03/04)

5. Operator vs. Automated Tasks

Step	Who	Why
1, 2	Automated (PowerCLI)	API operation on vCenter
3	Either UI or PowerCLI	vCenter UI gives best progress feedback
4	Automated (esxcli via PowerCLI)	API operation on ESXi
5	Operator (Workstation)	Workstation operation — outside vCenter
6	Operator (text editor on D:)	D: drive not accessible from automation context — manual edit required
7	Operator (Workstation)	Workstation operation
8-13	Automated (PowerCLI / esxcli)	API operations
14	Automated (PowerCLI poll)	Read-only verification

6. Rollback Procedure

If Steps 8 or 11 verification fails after the `.vmx` edit and disk-group recreate:

1. Power off esxi01 nested VM
2. Revert `.vmx` — remove the `sata0:4.virtualSSD = "1"` line
3. Power on esxi01
4. SSH and verify host stable
5. Exit MM
6. Disable Retreat Mode (Step 13)
7. Disk group can be left as-is (still hybrid, but functional)

The disk-group remove (Step 4) is permanent for that disk group UUID; rollback yields a new disk group with the same disks but different UUID, classifications matching the original (sata0:4 lsSSD=false). Host returns to pre-fix operational state.

7. Document Information

Field	Value
Document Title	vCLS Retreat + MM Retry + esxi01 Disk Fix

Field	Value
Date	May 15, 2026
Source MM failure event	2026-05-14 22:20-22:30 UTC
Disk group UUID (to remove)	521f8f5c-9912-c37b-461d-7b0b94e01e0c
Cluster ID	ClusterComputeResource-domain-c10
Retreat Mode advanced setting key	config.vcls.clusters.domain-c10.enabled
Classification	Internal — Lab Operations